

# COLX 585: Text-only and VLM Membership Inference Attack: Milestone 7

Yucai Zhong and Christina McCallum and Jennifer Flake and Rachelle De Jager

## Abstract

Membership Inference Attacks (MIAs) query a target model to determine if a particular data point was used in training. We applied various MIA methods to fine-tuned Large Language Models (LLMs) and Vision Language Models (VLMs) in a grey-box setting. For LLMs, we included features such as loss and perplexity, Min-K% and hard tokens signals. For VLMs, we employed temperature sensitivity, Max-Renyi-K%, neighbourhood comparison, and perturbation-based calibration across image and caption inputs, among other features.

Our primary contribution is cross-modal implementation and combination of recently-proposed MIA feature families, including methods aligned with the text-only setting extended to the VLM setting, and feature ablation to reduce signal dilution. Our best models achieved an AUC of 0.951 in the text-only setting and 0.8364 in the multimodal setting, suggesting that targeted feature combinations outperform simple uncertainty baselines.

## 1 Introduction

With the recent advancement and wide-reaching adoption of LLMs and VLMs, there are implications for data privacy across various domains including healthcare, finance, legal, and education. The presence of sensitive patient records, financial data and confidential legal documents in training sets indicates a need for strong privacy measures and compliance with jurisdiction-based privacy laws and regulations. Further, web-scraped training data raises concerns regarding use of copyrighted material and other personal data without consent. Previous research positions MIA methods as a critical technique in auditing these privacy risks, querying a model to establish whether a data point was used in training.

There has been limited work focusing specifically on fine-tuned VLMs and cross-modal comparison. VLMs are a newer development and more

complex than LLMs, which makes MIA methods more challenging to formulate given diverse signals from text and images. Our work explores these areas in a grey-box setting to find combinations of previously proposed feature-families.

Our main contributions are as follows: i) We applied a cross-modal extension of the neighbourhood comparison, originally proposed for LLMs by Mattern et al. (2023). ii) We proposed novel combinations of features sourced from individually published methods. iii) We directly compared feature sets in the text-only setting against the VLM setting. iv) We demonstrated that feature ablation outperformed adding and keeping more features. v) Our work showed that comparative signals from a fine-tuned model to a reference model outperformed models with features based on absolute metrics.

The remainder of this paper is organized as follows. Section 2 reviews related work on MIA methods for LLMs and VLMs. Section 3 describes the datasets and models used in both the text-only and multi-modal settings. Section 4 outlines our methods including feature derivation and classifier implementation. Section 5 details our experiments across all milestones. Section 6 presents our results using AUC-ROC and TPR@FPR=0.1 metrics. Section 7 concludes with a discussion of key findings and directions for future work.

## 2 Related Work

The overview of MIA on large language models and large multimodal models done by Wu and Cao in “Membership Inference Attacks on Large-Scale Models” (2025) is the foundation from which we built our initial features to explore what signals would be useful in detecting membership data. The authors discuss prominent types of attack strategies: loss-based and perplexity-based methods, neighbourhood/perturbation-based

approaches, Min-K% and Min-K%++, reference model methods, RAG systems, and model alignment data. Based on these ideas, we extracted features in milestones 1 and 2 like perplexity, loss, and Min-K%, and found improvement with the latter feature compared to our baseline (AUC of 0.8955 with XGBoost on the validation set compared to 0.8852 with logistic regression). Wu and Cao mention studies that use logistic regression, random forest, and neural network binary classifiers; we experiment with these models throughout our study to determine the best performing at each milestone.

More recent work by Jawad et al. in January 2026, “What Hard Tokens Reveal: Exploiting Low-confidence Tokens for Membership Inference Attacks against Large Language Models”, proposes the use of “harder” tokens, that is, tokens with the lowest confidence, which provide a much stronger signal of membership. Harder tokens will reveal systematically higher probabilities from the target model compared to the reference model. The authors use clinical notes datasets in their study and therefore provide an important MIA strategy in the same domain in which we operate. They achieve a highest AUC of 0.8843 on this dataset with the Qwen-3-0.6B model. In our method, combining the abovementioned features with HT-MIA (hard token features), we achieve 0.9511 on the test set. This is our highest performing model, though it’s important to note that we used different data and target models in our experiments compared to Jawad et al. We further show that their method achieves superior results and expand on it by combining existing features outlined above.

In the VLM paradigm, we extend Hu et al.’s 2025 work in “Membership Inference Attacks against Vision Language Models” and Li et al.’s 2024 work in “Membership Inference Attack against Large Vision-Language Models”. In milestone 5, we used a temperature-based attack and a MaxRenyi-K% feature. Hu et al. theorize that the temperature parameter can be used to control how random or confident the model is, and that members will be more sensitive to temperature parameter changes than non-members. Adding caption loss features at different temperatures, we saw an improvement from the baseline from AUC=0.7793 to 0.7846 on the validation set. MaxRenyi-K% and ModRenyi features, when added on top of the temperature features, only added small improvements to the model. Li et al. propose MaxRenyi-K% and the target-based version ModRenyi to measure the model’s

confidence in predicting each token based on Renyi entropy of the next-token probability distribution. Analyzing our top features, we found Renyi features did not add much signal beyond Shannon entropy and Min-K%, and that the temperature-based attack was more effective in adding membership signal by comparison.

Using the strategies in Mattern et al.’s “Membership Inference Attacks against Language Models via Neighbourhood Comparison” (2023), we created neighbour-based features to compare a sample’s likelihood to that of samples of neighbouring captions: members will receive substantially lower loss (and therefore higher likelihood) compared to neighbours. This achieved significant improvement compared to the temperature-based features, at AUC=0.8313 on the test set. Our approach is an important contribution to the field, as it applies a method originally proposed in the LLM setting to VLMs. Adding PC-MMIA proposed by Zheng et al. in “Tracing Training Footprints: A Calibration Approach for Membership Inference Attacks Against Multimodal Large Language Models”, we achieved our best score in the VLM domain with AUC=0.8364 on the test set. This perturbation-based calibration method measures the difference in token log-probability before and after image perturbation (high difference means the model saw the image-caption). By comparing these two approaches, we highlight feature importances in the VLM MIA domain.

Our research provides a systematic empirical comparison of feature families for MIA across LLM and VLM settings, highlighting which features make the most difference in which settings, and proposes novel combinations of existing work.

### 3 Datasets

In the first MIA task from milestones 1-3, we performed analysis on the SmolLM2-135M and SmolLM2-360M language models, which were both fine-tuned using patient discharge notes. The data was the UBC-SLIME/colx\_531\_group\_project dataset, with the target value is\_member labelled 1 if a sample was part of the fine-tuning data or 0 if it was not. The training set had 50,000 samples, the validation set had 10,000, and the test set had 15,000 (with no labels, as this is the prediction target). The training and validation splits are balanced, meaning that they contain approximately equal numbers of mem-

ber and non-member samples. Each example in the data contained the fields id (a unique identifier), text (the clinical text itself), and is\_member (the membership label). Within text, there is further metadata about the patient, including gender, age, and admission date, among others.

The text field is evenly distributed across the train/val/test split, with an average length of 1,950 characters. Among members and non-members, there is a similar number of text uniqueness, as demonstrated in Table 1.

Table 1: Text Uniqueness Analysis

| Metric                | Member            | Non-Member        |
|-----------------------|-------------------|-------------------|
| Duplicates            | 594/25k (2.4%)    | 667/25k (2.7%)    |
| Vocab. Richness (TTR) | $0.576 \pm 0.071$ | $0.577 \pm 0.071$ |
| Rare Content (TF-IDF) | $0.278 \pm 0.073$ | $0.279 \pm 0.074$ |

We also found n-gram rarity across 2-, 3-, and 4-grams to be relatively similar across member and non-member texts.

The second MIA task from milestones 4-6 involved the UBC-SLIME/colx\_858\_vlm model fine-tuned on image-caption pairs. The task was similar, but this time in the multimodal domain, and not with clinical data. The data, from UBC-SLIME/colx585\_group\_project\_data, contained 6,000 training samples, 1,200 validation samples, and 6,000 test samples. Each example in training and validation included id, image, is\_member, type, and text. Type indicates whether the target VLM has seen both the text and image, just the image, just the text, or neither the image nor the text. A sample is labelled is\_member = 1 if the target VLM has seen both image and text; all other types are labelled is\_member = 0 (i.e., there are multiple combinations of the 0 class). The train set has 1,500 members and 4,500 non-members while the validation set has 300 members and 900 non-members. Texts average 225-230 characters across all three splits, and all image sizes are the same. Consistency in lengths in both tasks meant that there was no length bias present to influence the MIA classifier.

In both cases, the input to the model is the training split, and the output is an extracted feature; all extracted features are input to a classifier, which outputs a membership score (AUC and TPR@FPR=0.1).

## 4 Methods

We formulated membership inference as a supervised binary classification task. Given an input example, the goal was to predict whether it was included in the fine-tuning data of the target model. The attack outputs a membership score, where a higher score means the example is more likely to be a member. We did not fine-tune the target models ourselves. Instead, we used the provided fine-tuned models and their corresponding reference models as frozen feature extractors. We first computed model-behavior features from each input, and then trained a downstream classifier on these features.

### 4.1 Text-only setting

For the text-only setting, we used two causal language models: UBC-SLIME/colx\_531\_smollm2-135m and UBC-SLIME/colx\_531\_smollm2-360m. Each clinical note was tokenized with a maximum length of 512 tokens and passed through both models to obtain next-token logits. From these outputs, we extracted confidence and uncertainty-related features, including loss, perplexity, token probability statistics, entropy-based measures, top-k mass, low-confidence token ratio, and Min-K% token probability features.

In later milestones, we added several stronger feature families, including Min-K++, HT-MIA scores, token-level log-likelihood ratio statistics, and zlib compression scores. We also constructed cross-model features by taking the difference and ratio between the 360M and 135M model features. These features were intended to capture relative confidence patterns across models of different sizes.

### 4.2 VLM setting

For the VLM setting, we used UBC-SLIME/colx\_585\_vlm as the fine-tuned target model and HuggingFaceTB/SmolVLM-Base as the reference model. Each example consists of an image-text pair. We split the input into a user prompt and an assistant caption, and computed loss only on the assistant caption by masking prompt and padding tokens.

From the target and reference VLM outputs, we extracted caption-level loss, perplexity, caption length, token probability statistics, Min-K% features, and entropy-based features. We also com-

puted comparative features between the target and reference models, such as caption loss difference, loss ratio, perplexity ratio, probability difference, and entropy difference. The expectation was that member samples should receive lower loss and sharper output distributions under the fine-tuned target model than under the base model.

### 4.3 Extended VLM features

We further explored several additional feature families in the VLM setting.

First, we added temperature-sensitivity features by rescaling logits with temperatures 0.25, 0.5, 2.0, and 4.0, and recomputing token probabilities and losses. From these values, we derived sensitivity-based features that reflect how the output distribution changes under temperature scaling. Second, we added Rényi-entropy features with  $\alpha = 0.5$  and  $\alpha = 2.0$ . We used both average Rényi entropy across caption tokens and top-K variants that focus on high-entropy positions. Third, we tested image-corruption features by applying Gaussian blur to the image and comparing the vision encoder outputs between original and corrupted inputs using patch-level cosine similarity statistics. Fourth, we introduced neighbor features by comparing the loss of the gold caption against the losses of 15 alternative captions for the same image and prompt context. These features include loss delta, loss ratio, percentile rank, and the variance of neighbor-caption losses. Finally, in the last milestone, we added PC-MMIA-style perturbation features, which measure calibrated changes in token log-probability after perturbing the image or text input.

### 4.4 Downstream classifiers and evaluation

For the downstream attack models, we evaluated Logistic Regression, Random Forest, Gradient Boosting, and XGBoost across different milestones. Logistic Regression served as the main linear baseline, while XGBoost was the strongest tree-based model in most experiments. We also tested a stacking MLP in milestone 6. Instead of using the full feature matrix directly, this model took the out-of-fold prediction scores from Logistic Regression and XGBoost as input and used a small multilayer perceptron to produce the final membership probability. We evaluated all methods using validation AUC and TPR@FPR=0.1. We also generated Kaggle-format predictions for the hidden-label test sets to measure external performance.

## 5 Experiments

We evaluated our methods across both the text-only and VLM settings. In each milestone, we tested different feature sets and downstream classifiers, and compared them using validation AUC, TPR@FPR=0.1, and Kaggle leaderboard performance. Below, we summarize the main results in each setting. The results suggested that membership inference can be highly effective in the text-only setting when the model-behavior features are well designed. In the multimodal setting, the same overall concept still provided some benefit, but the signal was weaker and careful feature selection became more important.

### 5.1 Text-only experiments

In milestone 1, we established a text-only baseline using 18 features extracted from the 135M and 360M models. We trained Logistic Regression and Random Forest classifiers. Logistic Regression performed best, with a validation AUC of 0.8852 and a Kaggle AUC of 0.89291. This result shows that even simple confidence-based features already provide a strong membership signal.

In milestone 2, we explored three extensions to the text-only pipeline. The first added Min-K% features and cross-model difference and ratio features. With this feature set, XGBoost achieved the best result, reaching a validation AUC of 0.8955 and a Kaggle AUC of 0.90548. The second experiment used cross-model delta features with Gradient Boosting and reached a validation AUC of 0.8905. The third experiment introduced partial SPV-MIA features, but this approach was limited by high extraction cost and incomplete feature coverage. Overall, milestone 2 showed that Min-K-style features and cross-model comparisons improved over the initial baseline.

In milestone 3, we further expanded the text-only feature set with Min-K++, HT-MIA, token-level log-likelihood ratio features, zlib compression scores, and a weighted soft-voting ensemble. This milestone gave the strongest text-only results. The weighted ensemble achieved a validation AUC of 0.9487 and TPR@FPR=0.1 of 0.8310. XGBoost also performed strongly with a validation AUC of 0.9470. In contrast, the zlib-only baseline was close to random, suggesting that model-behavior features were much more informative than compression features alone.

## 5.2 VLM experiments

In milestone 4, we moved from the text-only setting to the VLM setting. The baseline used caption loss, perplexity, token probability statistics, entropy, Min-K features, and target-reference comparative features. Among Logistic Regression, XGBoost, and a simple average ensemble, XGBoost performed best, with a validation AUC of 0.7793 and a Kaggle AUC of 0.76850. The strongest individual signals were caption loss difference and perplexity ratio. This suggests that target-reference caption likelihood differences are the main membership signal in the multimodal setting.

In milestone 5, we tested three VLM feature extensions. The temperature-sensitivity features gave a small improvement over the milestone 4 baseline, with the best ensemble reaching a validation AUC of 0.7846. The Rényi-entropy features performed slightly better, with the best ensemble reaching a validation AUC of 0.7879. The image-corruption features did not provide meaningful gains, and their performance stayed close to the baseline. These results suggest that likelihood-based and entropy-based caption features were more useful than image-only perturbation features for this task.

In milestone 6, we focused on feature merging, feature selection, and stronger ensemble design. In the first experiment, we combined temperature, Rényi, and image-perturbation feature families, added ID-based structural features, and selected a compact subset of informative features. Single-family feature sets already performed well, with Rényi features reaching a validation AUC of 0.8533. The combined global XGBoost model achieved a validation AUC of 0.8514. This suggests that careful feature filtering and consolidation were more helpful than simply adding more raw features.

In the second milestone 6 experiment, we added neighbor features and compared routed blending with neural stacking. The routed blend achieved a validation AUC of 0.8444, while the stacking MLP reached 0.8216. Although the MLP was more flexible, it did not outperform the simpler tree-based and routed approaches in this setting.

In the final milestone 6 experiment, we added PC-MMIA-style perturbation features together with pair-type and ID-based feature engineering. Pair-type routing improved validation performance to 0.8482, and adding PC-MMIA features with

feature selection reached the best validation AUC in the VLM setting, 0.8533. However, the corresponding Kaggle test AUC dropped to 0.83575, which suggested some validation overfitting or a mismatch between the validation and test distributions.

## 6 Results

### 6.1 Clinical Text MIA (Milestones 1–3)

For the first three milestones, membership inference was performed on clinical discharge notes. Results were reported using AUC and TPR@FPR=0.1. Validation AUC is reported for all models, with Kaggle test set scores for each milestone noted separately when available.

Performance improved across the three milestones. The baseline model included a set of nine features extracted from each of the two language models, for a total of 18 features. The baseline features included perplexity, loss, mean, minimum, maximum, and standard deviation of token probability, mean entropy, top-k mass, and low-confidence token ratio. Logistic regression and random forest classifiers were evaluated on these features, with logistic regression performing marginally better at validation AUC = 0.8852 and TPR@FPR=0.1 = 0.6492. These baseline features established that even simple output-level statistics from two models of different sizes carry a meaningful membership signal. The next major improvement came from the introduction of Hard Token MIA (HT-MIA) and log-likelihood ratio (LLR) features in Milestone 3. Table 2 reports validation performance and Table 3 summarizes Kaggle test performance for each milestone.

Table 2: Validation Performance: Clinical Text MIA (Milestones 1 to 3)

| Method                               | AUC           | TPR@FPR=0.1   |
|--------------------------------------|---------------|---------------|
| 1 Baseline: Logistic Regression      | 0.8852        | 0.6492        |
| 1 Baseline: Random Forest            | 0.8842        | 0.6554        |
| 2 Min-K% + XGBoost                   | 0.8955        | 0.6688        |
| 2 Cross-Model Delta + Gradient Boost | 0.8905        | 0.6496        |
| 2 SPV-MIA, Min-K%, XGBoost           | 0.8955        | 0.6662        |
| 3 Logistic Regression                | 0.9336        | 0.8272        |
| 3 Random Forest                      | 0.9399        | 0.8284        |
| 3 XGBoost                            | 0.9470        | 0.8302        |
| 3 Gradient Boosting                  | 0.9447        | 0.8232        |
| 3 Weighted Soft-Voting Ensemble      | <b>0.9487</b> | <b>0.8310</b> |

While the features introduced in milestone 2 modestly improved the metrics of the baseline, the

Table 3: Kaggle Test Performance: Clinical Text MIA (Milestones 1 to 3)

| Method                           | Kaggle AUC     |
|----------------------------------|----------------|
| 1 Baseline (Logistic Regression) | 0.89291        |
| 2 Min-K%, SPV-MIA, XGBoost       | 0.90549        |
| 3 HT-MIA + LLR + Ensemble        | <b>0.95105</b> |

LLR and HT-MIA features introduced in milestone 3 had more of an impact. By focusing on the hardest tokens in a sequence and measuring how much more confident the fine-tuned model was relative to the reference model at those positions, these features captured an improved memorization signal. Similarly, cross-model comparison features, such as delta and ratio features between the larger and smaller model were also useful. Relative features, instead of absolute model behavior, provided the more important learning signal. Lastly, the weighted soft-voting ensemble in milestone 3 provided another gain over any individual classifier.

## 6.2 VLM MIA (Milestones 4 to 6)

The VLM MIA part of the project proved more challenging than the clinical text phase, with baseline validation AUC starting at 0.7793 in milestone 4 and reaching a high of 0.8533 in Milestone 6. Table 4 reports validation performance and Table 5 reports Kaggle test performance across all models.

Table 4: Validation Performance: VLM MIA (Milestones 4 to 6)

| Method                           | AUC           | TPR@FPR=0.1 |
|----------------------------------|---------------|-------------|
| 4 Baseline: Log. Reg.            | 0.7760        | 0.2800      |
| 4 Baseline: XGBoost              | 0.7793        | 0.2833      |
| 4 Baseline: Ensemble             | 0.7790        | 0.2600      |
| 5 Temperature Feats: Log. Reg.   | 0.7841        | 0.2700      |
| 5 Temperature Feats: XGBoost     | 0.7816        | 0.2833      |
| 5 Temperature Feats: Ensemble    | 0.7846        | 0.2767      |
| 5 Rényi Features: Ensemble       | 0.7879        | 0.2867      |
| 5 Image Corruption: Ensemble     | 0.7819        | 0.2533      |
| 6 Temperature Feats + XGBoost    | 0.8495        | 0.3600      |
| 6 Rényi Features + XGBoost       | 0.8533        | 0.3400      |
| 6 Image Perturb + XGBoost        | 0.8455        | 0.3467      |
| 6 Global XGBoost (all feats)     | 0.8514        | 0.3133      |
| 6 Source-Routed XGBoost          | 0.8427        | 0.3133      |
| 6 Weighted Blend                 | 0.8330        | 0.2633      |
| 6 Neighbor Feats: Routed Blend   | 0.8444        | 0.2833      |
| 6 Neighbor Feats: Neural Stack   | 0.8216        | 0.2167      |
| 6 PC-MMIA (80 feats)             | <b>0.8533</b> | 0.2681      |
| 6 PC-MMIA (50 feats, lower corr) | 0.8514        | 0.2612      |

The single biggest gain in the VLM part of the project came from thoughtfully organizing existing features. The milestone 6 lightweight pipeline

Table 5: Kaggle Test Performance: VLM MIA (Milestones 4 to 6)

| Method                                  | Kaggle AUC    |
|-----------------------------------------|---------------|
| 4 Baseline (XGBoost)                    | 0.7685        |
| 5 Temperature Features (Ensemble)       | 0.7783        |
| 5 Rényi Features (Ensemble)             | 0.7804        |
| 5 Image Corruption (Ensemble)           | 0.7763        |
| 6 Lightweight Pipeline (Global XGBoost) | <b>0.8364</b> |
| 6 Neighbor Features (Routed Blend)      | 0.8318        |
| 6 PC-MMIA (80 features)                 | 0.8358        |

merged the three milestone 5 feature families, removed redundant features, and added lightweight structural ID features. This combination of feature filtering and adding structure produced an increase of approximately 0.057 AUC points on the Kaggle submission result, compared to the best milestone 5 result. Other gains throughout the VLM phase came from source-aware routing, which split samples by their dataset origin (cauldron vs. mix). In addition, in milestone 5, temperature-sensitivity features provided the most consistent gains in that milestone.

## 7 Conclusion

During the six milestones, we moved from a simple probabilistic baseline model for patient discharge notes to increasingly sophisticated models that used comparative features for image-text pairs. In milestones 1–3, we worked on clinical discharge notes with text-only language models. By milestone 3, the weighted ensemble reached a Kaggle AUC of 0.95105. During milestones 4–6, we switched to a more difficult problem: multimodal MIA on a small VLM (SmolVLM-256M), where the best Kaggle AUC we could achieve was 0.83480.

The two tasks required different methods. Many of the probabilities successfully used as features in the text MIA models did little to improve the models for the VLM MIA task. Furthermore, many of the image-based features we tried in the VLM MIA task, including corruption, PC-MMIA and perturbation never added learning signal. While specific implementations and goals varied between the two tasks, we can draw some broader conclusions about the entire project.

For both the text only task and the VLM task, our baseline models included more than simple baseline features. While going beyond the typical minimalist baseline was not necessary, we found that doing so allowed us to perform more experiments with features and techniques in the milestones that

followed.

In both phases, compute and time constraints had a major impact on what was possible to implement, train, and improve during a milestone. For example, the models that contained image-related features took approximately 12 hours to train. With so many hours spent training, there was not much time to iterate on features after a training run. Furthermore, we often had to limit the amount of data we used for training or limit the implementation of features to decrease the amount of time required in training. Unfortunately, some of the features, like PC-MMIA and perturbation, that we read about in literature could not be implemented in an effective way. If we had access to more time and more compute, perhaps our model could have realized the gains that the literature discussed.

In both phases of the project, we found that comparative features were superior to features using absolute metrics. For example, models with features that compare a fine-tuned model to reference (base) model outperformed models that only used absolute metrics like raw loss or perplexity. Similarly, in the text-only phase of the project, features that focused on differences between the SmolLM2-135M and SmolLM2-360M language models were effective capturing valuable learning signals.

By trying combinations of many features, we learned that adding more features does not necessarily improve a model. In fact, weak and redundant features dilute the learning signal. In Milestones 2 through 5, we kept adding new feature families, including Min-K%, Rényi entropy, temperature sensitivity, image corruption, and Rényi-based variants. Each time new features were added, the gains were small. In milestone 6, when we systematically filtered features, instead of adding more, we got the biggest single milestone jump in validation AUC. Systematic feature selection beat complex feature engineering.

Given the opportunity to do a similar project, our team would do some things differently. For example, we would start with more modular, reusable code during milestone 1. We would save extracted features to disk for reuse from the start. By milestone 3, we had increased modularity, by using a slim driver notebook and modular python script files. At that point, we started saving extracted features to disk for reuse. Moreover, we would approach the addition of new features and the ablation of weak or redundant features in a more methodical way. Finally, we would perform proper hyperpa-

rameter searches from the start rather than relying on defaults.

## 8 Generative AI Usage

While the team developed the overall methods and architecture of our models, generative AI was used extensively in the low level code implementation details. Many, if not all, of the methods that we employed are informed by research papers. Given the complexity of the features suggested by the papers, as well as the time and compute constraints of each milestone, assistance from Anthropic’s Claude for programming details was necessary to meet weekly deadlines.

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